Aditya Mittha

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Summary

Final-year engineering student with a solid foundation in Python development and hands-on DevOps practices. Experienced in writing Python scripts for automation, data processing, and background services, alongside building serverless backends on AWS. Practical experience containerizing applications with Docker, setting up CI/CD workflows using GitHub Actions, and managing Linux environments. Eager to leverage Python programming and cloud automation skills in an entry-level Python / DevOps role.

Skills

Programming & Scripting: Python (Data Structures, OOP, REST APIs, Automation), Bash, C, JavaScript

DevOps & Containerization: Docker, Dockerfile creation, Container builds, CI/CD Concepts

Cloud Platforms (AWS): AWS Lambda, API Gateway, DynamoDB, S3, IAM, CloudWatch, Firebase

CI/CD & Tools: Git, GitHub, GitHub Actions (Automated Image Builds), Postman, JSON/YAML

Operating Systems & Networking: Linux (Ubuntu/WSL, Shell Commands, SSH), Windows, TCP/IP, HTTP/HTTPS, DNS

Soft Skills: Problem Solving, Team Collaboration, Adaptability, Clear Communication, Self-Learning

Education

Bachelor's Degree in Electronics and Telecommunication 2027
Walchand Institute Of Technology *CGPA: 9.27*

Higher Schooling 2023
D.H.B Soni College, Solapur *Percentage: 83%*

Work Experience

Embedded Systems Intern May 2025 – June 2025
Codec Technologies India *Remote*

- Set up and tested UART and I^2C communication interfaces on microcontrollers, writing C code to establish reliable hardware-to-software data transmission.
- Debugged hardware interfacing issues across two embedded modules, resolving 10+ integration defects prior to production testing.
- Utilized Git and GitHub for version control across active branches, maintaining structured commits and up-to-date technical documentation.
- Worked in a Linux terminal environment for firmware flashing, serial log analysis, and executing test automation scripts.

Technologies: Python Scripting, C, UART, I^2C, Git, GitHub, Linux, Serial Debugging

Projects

LabPulse – Cloud-Based Lab Monitoring System 2026

- Developed a modular Python agent running as a Windows background service to track active user sessions and securely stream telemetry to the cloud API.
- Designed and deployed the AWS serverless backend (Lambda, API Gateway, DynamoDB) to ingest, process, and store system usage metrics asynchronously.
- Containerized the dashboard application using Docker and configured a GitHub Actions CI/CD pipeline to automate Docker image builds on code commits.
- Implemented API key authentication with SHA-256 hashing and Role-Based Access Control (RBAC) to secure endpoint interactions.

Role: Backend Lead, Cloud & DevOps (3-person team)

Tech Stack: Python, AWS Lambda, API Gateway, DynamoDB, Docker, GitHub Actions, React, PowerShell

Air Guitar – Real-Time Sensor-to-Audio Pipeline 2025

- Engineered a real-time Python application that ingests MPU6050 accelerometer data from an Arduino via serial communication and generates spatial audio.
- Developed signal processing algorithms using Python libraries (NumPy, sounddevice) to map physical sensor angles to virtual string triggers via the Karplus-Strong algorithm.
- Designed clean Object-Oriented (OOP) abstractions separating data acquisition, mathematical calculations, and audio rendering modules.

Role: Solo Project — Python Development, System Design

Tech Stack: Python, NumPy, OOP, I^2C, Serial Communication, Hardware Interfacing

Rewards & Achievements

- 2x Winner, National Electronic Quiz Competition (Quizotronics)
- Runner-up, State-level chess championship
- Participated in 4^+ Hackathons

Languages

English | Hindi | Marathi | Kannada